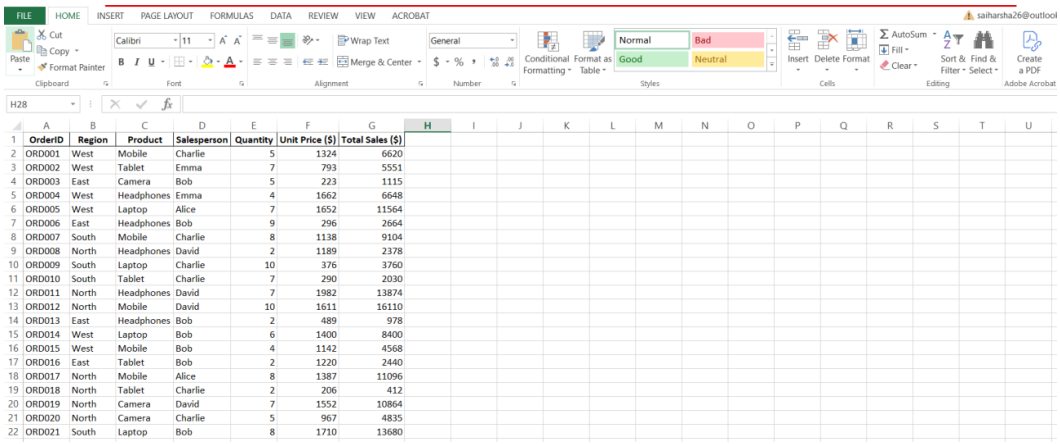
Advanced Excel

Agenda

- Introduction to Excel and Excel Interface
- Advanced Formulas & Functions
- Conditional Formatting
- Data Cleaning Tools
- PivotTables & PivotCharts
- Data Analysis Tools
- Advanced Charts
- Case Study
- Tips & Best Practices

Introduction to Excel and Excel Interface

- MS Excel is a commonly used Microsoft Office application.
- It is a spreadsheet software which is used to save and analyze numerical data.



OrderID	Region	Product	Salesperson	Quantity	Unit Price ($)	Total Sales ($)
ORD001	West	Mobile	Charlie	5	1324	6620
ORD002	West	Tablet	Emma	7	793	5551
ORD003	East	Camera	Bob	5	223	1115
ORD004	West	Headphones	Emma	4	1662	6648
ORD005	West	Laptop	Alice	7	1652	11564
ORD006	East	Headphones	Bob	9	296	2664
ORD007	South	Mobile	Charlie	8	1138	9104
ORD008	North	Headphones	David	2	1189	2378
ORD009	South	Laptop	Charlie	10	376	3760
ORD010	South	Tablet	Charlie	7	290	2030
ORD011	North	Headphones	David	7	1982	13874
ORD012	North	Mobile	David	10	1611	16110
ORD013	East	Headphones	Bob	2	489	978
ORD014	West	Laptop	Bob	6	1400	8400
ORD015	West	Mobile	Bob	4	1142	4568
ORD016	East	Tablet	Bob	2	1220	2440
ORD017	North	Mobile	Alice	8	1387	11096
ORD018	North	Tablet	Charlie	2	206	412
ORD019	North	Camera	David	7	1552	10864
ORD020	North	Camera	Charlie	5	967	4835
ORD021	South	Laptop	Bob	8	1710	13680

Advanced Formulas & Functions

- IF, Nested IF, COUNTIF
- Dynamic Array functions
- VLOOKUP, HLOOKUP, XLOOKUP

Conditional Formatting

- Color scales and icon sets
- Custom formulas for formatting
- Highlighting trends and outliers

Data Cleaning Tools

- Remove Duplicates
- Text-to-Columns
- Flash Fill for patterns

PivotTables & PivotCharts

- Create PivotTables
- Grouping, Filtering, Slicers
- PivotCharts for visualization

Data Analysis Tools

- Goal Seek
- Scenario Manager
- Data Tables

Advanced Charts

- Combo charts
- Sparklines
- Waterfall charts

Case Study

- Amazon Sales Data
- Cleaning dataset
- Using formulas for analysis
- Building PivotTables & charts

Tips & Best Practices

- Keep formulas simple & documented
- Use Tables for structured data
- Validate data before analysis
- Copy the raw data into another file then do your analysis on that copied file